

Answer all questions.

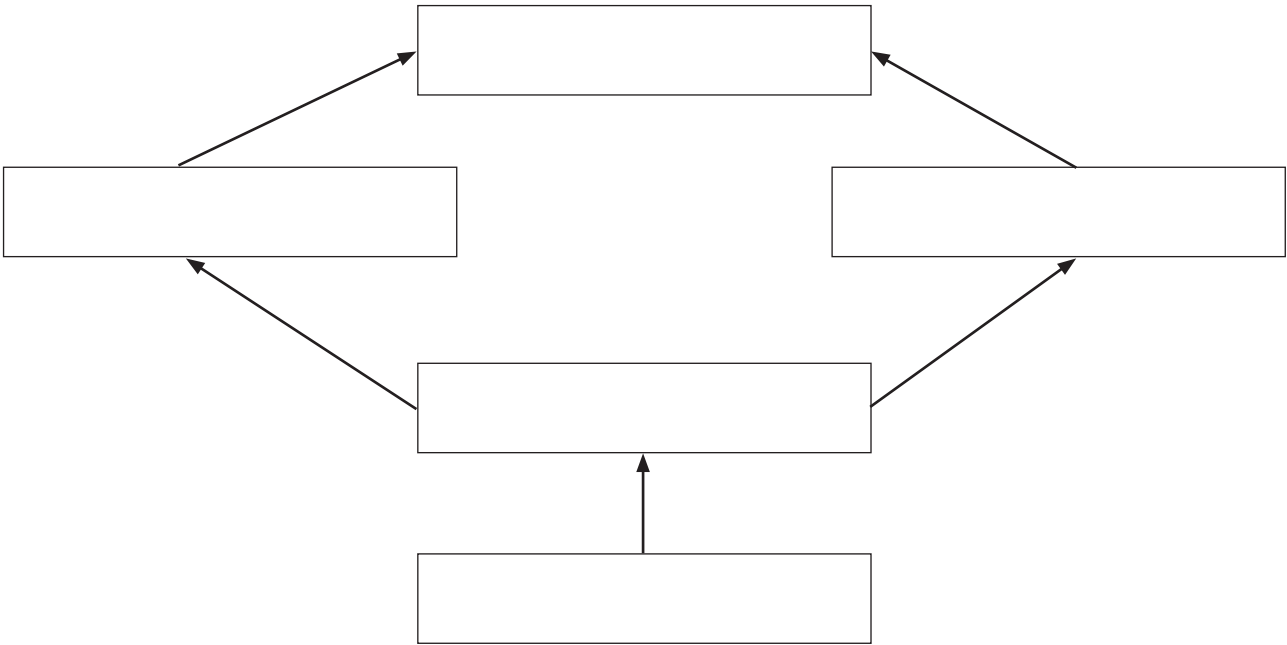
1. Killer whales (*Orcinus orca*) feed on thousands of large fish and squid. These large fish and squid feed on hundreds of small fish. Small fish feed on tiny organisms called plankton.

In 2016, a dead killer whale was found washed up on an island off the West Coast of Scotland. Tests revealed the whale contained one of the highest levels of PCBs (polychlorinated biphenyls) ever recorded. Levels of PCBs are measured in milligrams per kilogram of fat in an animal's body. Scientists believe that the threshold where PCBs can cause harm to killer whales is 30 mg/kg of fat. The level of PCB in the dead whale was 957 mg/kg of fat.

During the last century, PCBs were widely used in plastics, electrical goods and cement. PCBs were banned in the 1970s. It is estimated that there are a million tonnes of PCB-contaminated material waiting to be disposed of in Europe. PCBs are released into the environment from decomposing products that have not been disposed of properly. Plankton absorb PCBs that are washed into the oceans.

There are only eight resident killer whales remaining in UK waters. Scientists have not observed any offspring being born to the resident killer whales in the 25 years they have been studying them.

- (a) (i) Complete the simple food web below to show the feeding relationships of a killer whale. [3]



- (ii) State the term used to describe the feeding stage of killer whales. [1]

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- (b) Calculate how many times greater the level of PCBs was in the dead killer whale’s body when compared to the threshold where PCBs cause harm. [1]

Times greater =

- (c) Suggest the name of a group of micro-organisms responsible for releasing PCBs into the environment. [1]

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- (d) Explain how the PCBs led to the death of the killer whale. [2]

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- (e) Apart from causing death, describe another effect of PCBs on killer whales in UK waters. Give evidence to support your answer. [2]

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- (f) Suggest why PCBs are still present in UK waters even though they have been banned since the 1970s. [1]

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